A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 21

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1** Igneous intrusions (batholiths) of magma cool within the crust, heating groundwater to produce superheated, pressurised water **1** The hot water dissolves minerals from the surrounding rock and the mineral-rich solutions travel along fissures / cracks away from the batholith **1** As the solutions cool the dissolved minerals crystallise in order of solubility (least soluble first) - fractional crystallisation **1** So different metals are separated and concentrated in veins, eg tin, copper, lead, silver, gold, arsenic (SW England batholith) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Dissolved iron compounds in ancient seas were oxidised by the oxygen released by early photosynthetic organisms (2.5-1.8 billion years ago) **1** Iron oxides are insoluble so they were deposited as sediments (haematite, magnetite) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Flowing water carries solid particles – the ability to carry them depends on the water velocity and the density of the solids **1** As velocity falls, dense minerals (gold, tin ore) are deposited first while lighter material is carried further, concentrating the heavy minerals in particular places **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Limestone / chalk from the shells of marine organisms **1** Coal from terrestrial vegetation **1** Crude oil and natural gas from marine organisms (plankton) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 132-134; AQA spec 3.2.3.2.	
02.1	**1 mark each** 1 mark for each of the 4 points below. **1** Affluent people buy more consumer goods / trivia / disposable items and throw items away rather than repairing them **1 Built-in obsolescence:** products designed to have short lives / wear out / go out of fashion so customers upgrade **1** Disposable products (razors, lighters, pens) that must be replaced **1** Over-packaging to make products attractive **OR** increases wastes from manufacture and mineral extraction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
02.2	**Any 3 of 7** 1 mark each for any 3 of the 7 points below, in any order, maximum 3. **1 Properties of the waste:** toxicity, flammability, degradability, radioactivity **1** Population density (collection costs) **1** Land availability for landfill **1** Availability of recycling technology / markets **1** Legislation / regulatory framework (landfill tax) **1** Public willingness to recycle / environmental awareness **1** Processing costs / household income **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1 Advantage:** aerobic decomposition produces little methane – nutrients / organic matter returned to soil (circular economy) – reduces landfill volume **1 Disadvantage:** needs separation / collection of clean organic waste – odour / pests / pathogens if poorly managed – low-value product – not suitable for mixed waste **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
02.4	1 mark for the point below. **1** Many different materials are mixed together / contaminated so identification and separation are costly and the quality of recovered material is reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-282; AQA spec 3.4.3.2.12.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1 Conventional:** 40 / 100 = 40% **1 CHP:** (35 + 45) / 100 = 80% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **CHP** → using the waste heat from generation for heating **Do not credit** • A bare wrong answer with no working.	2
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Extracting less energy as electricity leaves the cooling / condenser water at a higher temperature **1** Hotter water is more useful for district heating of homes, fish farms, greenhouses **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CHP** → using the waste heat from generation for heating	2
03.3	1 mark for the point below. **1** Hot water loses heat and needs pumping energy over distance – heat cannot be transported as far as electricity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CHP** → using the waste heat from generation for heating	1
03.4	**1 mark each** 1 mark for each of the 3 points below. **1** Long narrow pipes give a large surface area and copper is a good thermal conductor so heat transfers quickly **1** Hot and cold fluids flowing in opposite directions maintain a temperature gradient along the whole length so more heat is transferred **1** Slower flow allows more time for heat to transfer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
03.5	**1 mark each** 1 mark for each of the 2 points below. **1** One large tank has a lower surface area : volume ratio than many small ones **1** A sphere has the smallest surface area for any volume – less surface means less heat loss (with insulation of low thermal conductivity) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 233-235; AQA spec 3.3.4.2.3.	
04.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 25 g ha^{-1} (accept 24-27) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (g ha^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
04.2	1 mark for the point below. **1** The highest dose that can be applied without killing any members of the population / any individuals of the (non-target) species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
04.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** The MDAF (minimum dose always fatal) of the pest is about 50 g ha^{-1} **1** This is higher than the MDNF of the bee (about 20 g ha^{-1}) **1** So a dose that kills all pests kills some bees / the two curves overlap between about 20 and 50 g ha^{-1} **1** A dose of 30-40 g ha^{-1} would kill most pests with limited bee deaths (compromise) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
04.4	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Spray at night / when bees are not flying **1** Spray when the crop is not in flower **1** Spray on still days to reduce spray drift **1** Use a systemic pesticide / seed coating rather than spraying **1** Use smaller droplets for more even coverage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
04.5	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Specificity is the extent to which the pesticide is toxic only to the target species / range of species affected **1** A highly specific pesticide has low toxicity to non-target species (eg bees, fish, mammals) **1** So fewer non-target deaths / less loss of predators, pollinators **1** Fewer indirect (food chain) effects / fewer inter-species relationships lost **1 Example:** pyrethroids have high insect toxicity but low mammal toxicity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-263; AQA spec 3.4.3.2.8.	
05.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Water = 65.0 - 55.0 = 10.0 g – fresh soil = 40.0 g **1** 10.0 / 40.0 x 100 = 25.0% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Organic matter = 55.0 - 52.6 = 2.4 g – dry soil = 30.0 g **1** 2.4 / 30.0 x 100 = 8.0% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** So all water is removed and the mass loss on burning is due to organic matter only **1** Heating until mass no longer changes shows all water has evaporated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.4	**1 mark each** 1 mark for each of the 3 points below. **1** Organic matter releases nutrients as it decomposes and provides food for soil biota **1** Increases water retention and, as humus, binds particles into crumb peds improving structure / aeration **1** Surface organic matter / humus adhesion reduce rain splash and wind erosion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **peds** → soil crumbs / aggregates of soil particles	3
05.5	1 mark for the point below. **1** Samples taken from different depths / locations (organic matter is concentrated near the surface) – incomplete burning – different heating times **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-159, 412; AQA spec 3.2.5.1, 3.7.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora **1** 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants) – 100 m x 100 m is for trees (impractical, too much work) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area	2
06.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** N = 240 – N(N-1) = 240 x 239 = 57 360 **1** Σn(n-1) = 48x47 + 120x119 + 36x35 + 24x23 + 12x11 = 2256 + 14280 + 1260 + 552 + 132 = 18 480 **1** D = 57 360 / 18 480 = 3.10 (accept 3.1) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Simpson** → a diversity index that uses both the number of species and their abundance • **Biodiversity** → the variety of life: habitat, species and genetic diversity **Do not credit** • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3
06.3	**1 mark each** 1 mark for each of the 2 points below. **1 Lower D means lower biodiversity:** the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) **1** Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **plagioclimax** → a community held back from its climax by human activity	2

Do not write outside the box

Question	Marking guidance	Marks
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants **1** Point quadrat pins are lowered through the vegetation – each touch counts (10 pins = 10% each) without flattening **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area	2
06.5	1 mark for the point below. **1** Subjective judgement varies between recorders – results are not quantitative so statistical analysis is not possible (but quick for many areas) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	
07.1	**1 mark each** 1 mark for each of the 5 points below. **1 Double hull:** a gap of up to 2 m between the hulls so damage to the outer hull in a collision / grounding does not release oil **1 Inert gas system:** cooled engine exhaust gas (no oxygen) fills the space above the oil so an explosive oil vapour / air mixture cannot form **1 Separate ballast tanks:** ballast water is not carried in oil tanks so no residual oil is discharged when ballast is pumped out **1 Twin engines / rudders:** if one fails the ship can still be steered / propelled and is not carried onto rocks by currents or wind **1 Crude oil washing:** sludges are mixed and removed with the cargo so tanks do not need washing with seawater that was then discharged – oily water is unloaded at the terminal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Offshore routes keep tankers further from the coast so if there is engine failure there is more time to deal with it before the ship is carried ashore **1** GPS gives accurate navigation – AIS lets ship positions be monitored and coordinated to avoid collisions and groundings **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	2

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** The tank stands on an impermeable base surrounded by a wall **1** The volume enclosed by the wall is greater than the tank volume so all the oil is contained if the tank splits / cannot overflow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through	2
07.4	1 mark for the point below. **1** Separates floating oil from drainage water so the oil is retained (for later treatment) and only water flows to the drains / river **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 259-260; AQA spec 3.4.3.2.7.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Tropospheric ozone (formed by reactions of NOx in sunlight)	1
08.2	**1 mark each** 1 mark for each of the 4 points below. **1 Primary pollutant:** released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores – NOx from high-temperature combustion **1 Secondary pollutant:** produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) **1** Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid – NOx dissolve to form nitric acid **1** The acids (sulfurous, sulfuric, nitric) are secondary pollutants – ozone itself is a secondary pollutant of NOx photochemistry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** A fluid in a buried ground loop absorbs low-temperature heat from the ground – at the expansion valve the fluid boils at low pressure absorbing this heat **1** A compressor raises the pressure so the gas condenses, releasing heat at a higher temperature inside the building **1** The electricity only moves heat rather than creating it, so up to four times as much heat is delivered as electricity used (less if the source is colder) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Parabolic reflectors focus sunlight onto tubes of oil, increasing the energy density and reaching high temperatures **1** The heat is transferred to molten salt (potassium nitrate) stored in large insulated tanks at up to 550 °C **1** The stored heat boils water to drive steam turbines whenever electricity is required, including at night **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reflectors can only focus parallel rays of direct sunlight – cloud, dust and smoke scatter light so it cannot be concentrated **1** PV cells convert diffuse as well as direct light, though output is lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **diffuse** → pollution from a wide area with no single outlet	2

Do not write outside the box

Question	Marking guidance	Marks
09.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1 Manufacture:** extraction / processing of silicon, metals and plastics – toxic wastes (silicon tetrachloride, cadmium) **1** Land take by large solar farms competing with other uses / affecting desert flora and fauna **1** Water for cleaning panels in arid areas **1** Disposal of old panels **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 200-202; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Control of greenhouse gases:** energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture • **Geoengineering:** white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences • **International agreements:** Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation – carbon trading and taxes – comparison of per capita carbon footprints between countries • **Evidence of limited success:** global CO2 concentrations still rising (about 280 ppm pre-industrial to over 410 ppm) – temperatures up about 1 °C – emissions in some developed countries falling but rising in developing economies • **Reasons for difficulty:** fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop • Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause • Comparison with the Montreal Protocol's success for ozone • **Judgement:** partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise – tipping points make urgency greater **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	9
	• Any accurate, relevant material that is not in the indicative content. • **energy conservation** → using less energy, eg by insulation or behaviour change • **CCS** → capturing carbon dioxide from flue gases and storing it underground **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	Strategies fall into control of greenhouse gases, geoengineering, international agreements and adaptation. Emissions are cut by energy conservation in transport, buildings and industry, by replacing fossil fuels with renewables and nuclear, by carbon capture and storage, by reducing deforestation and planting forests, and by capturing methane from landfill. Geoengineering ideas - white roofs, ocean fertilisation, solar shades - remain largely untried with unpredictable side effects. Adaptation, such as flood defences and managed retreat, reduces impacts but not the cause.	
	The Kyoto Protocol and Paris Agreement set targets, but they are weak or non-binding and participation is uneven; carbon trading and taxes exist but per capita footprints still differ enormously between countries. The evidence of success is limited: atmospheric CO_2 has risen from about 280 ppm before industrialisation to over 410 ppm and temperature by about 1 °C, and although emissions are falling in some developed countries they are rising in developing economies.	
	The difficulty is that fossil fuels underpin economies, so control demands lifestyle change, cost, and political agreement between nations with different priorities, while time delays and natural fluctuations mask the effect of any action. By contrast the Montreal Protocol succeeded because ozone was a substitution problem. The judgement must be partial success: the technology exists and regional reductions prove it works, but global strategies have not yet stopped the rise, and tipping points make the delay dangerous.	
	Weaker answer - Level 2 (5 marks)	
	Methods to reduce climate change include using renewable energy such as wind and solar instead of fossil fuels, saving energy, and planting trees to absorb carbon dioxide. International agreements like the Kyoto Protocol and the Paris Agreement have set targets for countries to reduce emissions. However carbon dioxide levels are still rising, so the methods have not been very successful so far. This is because countries rely on fossil fuels and do not always keep to the agreements. More needs to be done.	
	What makes the difference	
	• Level 2 covers the main strategies and reaches a conclusion, but with no quantitative evidence (280 to 410 ppm, 1 °C) and no mention of geoengineering or adaptation.	
	• Level 3 explains *why* success is limited (economic dependence, time lags, natural variability, political disagreement) rather than just stating that it is.	
	• The Montreal comparison and the tipping-point point give Level 3 the breadth and judgement the top band requires.	
10.2	1 mark for the point below. **1** The total greenhouse gas / CO2 emissions caused by an individual, organisation, product or country (often expressed per capita per year) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.	
11.1	**Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • **Fossil fuels:** choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports – new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • **Nuclear:** high energy density reduces transport – waste properties (half-life) determine storage – thorium and fusion prospects • **Renewables:** matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting) – intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity – low-impact designs (helical turbines, VAWT, tidal lagoons) • **Energy conservation reduces demand:** transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use	25
	– life-cycle / embodied energy • **Evaluation:** no resource is impact-free – knowledge allows the best mix and mitigation – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Context:** uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • **New sources:** rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • **Management:** artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • **Conservation:** low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering – pollution control to protect sources • **Treatment:** matching quality to use – screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	

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